

Philosophy of Science, September-October 2026

- 5354PHSC6Y in study guide, 6 EC
- Master of Logic, Master Physics and Astronomy
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- Tutorials: Luka Vinck (l.j.vinck@uva.nl), Matteo Fabbri (m.fabbri2@uva.nl).

Objectives

1. To acquire knowledge of central debates in contemporary philosophy of science at the master's level.
2. Skills: handling primary literature in philosophy of science; presenting a text in class, if necessary in collaboration with a fellow student; to constructively debate a text at an academic level; to be able to concisely represent a text in short writing assignments.
3. Attitude: to be able to critically reflect upon scientific knowledge from a meta-perspective; to be able to approach a text analytically.

Content

This course addresses central subjects in the contemporary philosophy of science, including:

- (1) The relation between theory and evidence: the logical empiricist approach to science, induction, prediction, and evidence.
- (2) Kuhn's philosophy of science.
- (3) The debate over scientific realism: the 'no-miracles argument', Van Fraassen's constructive empiricism, the pessimistic meta-induction, inference to the best explanation, models of selective confirmation, structural realism.
- (4) The Duhem-Quine thesis.
- (5) Explanation and understanding.
- (6) Laws of nature.

Forms of tuition

- Lectures
- Tutorials

Study materials

M. Curd and J.A. Cover, 'Philosophy of Science, the central issues', New York, Norton. You should have a copy of the book at the beginning of the course.

Other literature can be downloaded from the UvA library using the provided link, or will be made available on Canvas.

Assessment

- Paper (30%)
- Presentation (20%)
- Exam (50%). To pass the course, a minimum mark of 5.5 on the exam is required.

Weekly Programme

Week	Topic	Readings Unless otherwise indicated, the texts are from Curd & Cover. Page numbers are from the International Student Edition (2013) [1998 Edition]. Texts shown <u>underlined</u> can be downloaded from the UvA library. The remaining readings on Canvas are available under Files.
1 (36)	What is science? (Chapter 1) Logical empiricism	• Carnap, <i>The Elimination of Metaphysics</i> (on Canvas) • Godfrey-Smith, <i>Logic Plus Empiricism</i> (on Canvas) • Popper, <i>Science: Conjectures and Refutations</i> pp. 3-10 [3-10]
2 (37)	Kuhn on scientific progress (Chapter 2)	• Lakatos, <i>Science and Pseudoscience</i> pp. 20-26 [20-26] • Kuhn, <i>The Nature and Necessity of Scientific Revolutions</i> pp. 79-93 [86-101] • Longino, <i>Values and Objectivity</i> pp. 144-164 [170-191]
3 (38)	The Duhem-Quine thesis (Chapter 3)	• Quine, <i><u>On Empirically Equivalent Systems of the World</u></i> • Laudan, <i>Demystifying Underdetermination</i> pp. 288-320 [320-353] • <i>Optional: Duhem, <i>Physical Theory and Experiment</i></i> pp. 227-249 [257-279]
4 (39)	Explanation and understanding (Chapter 6)	• Godfrey-Smith, <i>Explanation</i> (on Canvas) • De Regt and Dieks, <i><u>A Contextual Approach to Scientific Understanding</u></i> • <i>Optional: De Jong and De Haro, <i><u>Technological Understanding</u></i></i>
5 (40)	Scientific realism (Chapter 9)	• Bas van Fraassen, <i>Arguments Concerning Scientific Realism</i> pp. 1060-1082 [1064-1087] • Musgrave, <i>Realism versus Constructive Empiricism</i> pp. 1083-1107 [1088-1113] • Laudan, <i>A Confutation of Convergent Realism</i> pp. 1108-1128 [1114-1135]
6 (41)	Scientific realism	• Psillos, <i>Resisting the Pessimistic Induction</i> (on Canvas) • Stanford, <i><u>'Atoms Exist' Is Probably True, And Other Facts That Should Not Comfort Scientific Realists</u></i> (only Sections 1-2.4, 3.1-3.2, 4.1-4.3) • <i>Optional: De Haro, <i>An Extensional Scientific Realism, (I): The Continuity of Extensions</i></i> (on Canvas)
7 (42)	Laws of nature	• Chen, <i><u>Laws of Physics</u></i> • <i>Optional: Carroll, <i><u>Laws of Nature</u></i></i>